

Derek Martin

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Professional Summary

Mechanical engineer with an aerospace focus and physics minor, combining hands-on mechanical design, Python automation and control development, instrumentation, and field engineering experience. Experienced with CAD-based design, mechanical analysis, SCADA systems, technical troubleshooting, and multidisciplinary project work.

Education

B.S. Mechanical Engineering, Aerospace Engineering Focus, Physics Minor

Oregon State University | Corvallis, Oregon

Graduated: September 2026

Master's in Engineering | Expected Graduation: June 2027

GPA: 3.68 | Dean's List

Relevant Coursework: Space Systems Engineering, Systems Dynamics & Control, Dynamics, Heat Transfer, Mechanics of Materials, Instrumentation

Professional Experience

Engineering Intern | Evensol Inc. - EDG-L (Landfill Gas Collection) | June 2024 - September 2024

- Conducted equipment inspections, troubleshooting, and preventative maintenance on critical systems.
- Designed and simulated a motor-maintenance hand-crank system in Autodesk Fusion 360, applying 3D modeling and engineering analysis to evaluate the design.
- Supported engineering feasibility studies, project pricing, cost estimates, and technical documentation.
- Developed and maintained Excel-based data-tracking tools for analysis, reporting, and operational decision-making.
- Supported SCADA systems, transmitter calibration, gas sampling, and instrumentation-related field work.
- Collaborated across engineering and field teams on equipment installations, troubleshooting, and infrastructure improvements in a safety-critical environment.

Wildland Firefighter | USFS Sisters Ranger District | August 2022 - September 2023

- Implemented fire suppression techniques, including direct and indirect fire line construction, demonstrating strong problem-solving under pressure.
- Operated various firefighting equipment and maintained chainsaw certification for hazardous tree removal and vegetation clearing.
- Collaborated effectively within a high-stakes team environment, utilizing navigation and GPS tools for efficient wildfire suppression.

Technical Skills

Software: Python (automation/control), Microsoft Excel, 3D Modeling (Autodesk Fusion 360), SCADA

Engineering Analysis: Stress analysis, structural loading, factor-of-safety evaluation

Safety Protocols: PPE, working at heights, emergency response

Project Management: Evaluation, scheduling, resource management, data analysis and documentation

Systems: Space systems fundamentals, instrumentation exposure, reliability & safety mindset

Certifications

Emergency Medical Technician (EMT) | Certified through Impact EMS | June 9, 2024

Achievements

- Awarded "Hardest Worker" in Calculus (2021)
- Team Captain for Nordic Skiing
- Medallion awarded by Bend Police Department for integrity and discipline (2021)

Projects

- Autonomous Piano Player (Capstone, 2025–2026): Developed Python control software for an electromechanical system and supported CAD, mechanical integration, and system testing.
- Winch Lift Stand: Designed and analyzed an aluminum stand supporting a winch to raise a 150 lb motor ~50 ft with a high factor of safety.
- Food Stirrer Prototype: Performed stress analysis and supported iterative design for a 3D-printed liquid-food stirrer with a 4-person team.